Optimization Models

Ulink MM Club

Frank

The Objective Function

The objective function is something you aim to maximize (or minimize), it represents the final goal of the model. For example:

- Shortest time;
- Lowest cost;
- Largest revenue;
- …

The Universal Form

$$
\begin{aligned}
&\text{minimize } f(x) \\
&\text{subject to } g_{\mathrm{i}}(x) \leq 0, i = 1,2,3, \cdots, m \\
&\qquad\qquad\quad h_{\mathrm{j}}(x) = 0, j = 1,2,3, \cdots, p \\
&\qquad\qquad\quad x \in \mathcal{X}
\end{aligned}
$$

where:

$f(x)$ denotes the objective function;

x denotes the decision variables;

$g_{\mathrm{i}}(x)$ and $h_{\mathrm{j}}(x)$ denote the constraints.

Linear Programming (LP)

The objective function and constraints are all **linear**.

$$\text{minimize } f(x) = c^T x$$
$$\text{subject to } Ax \leq B.$$

This is the simplest case of optimization models.

Linear Programming (LP)

You need to purchase apples of $x_1$kg and bananas of $x_2$kg. In total, you must but at least 4kg, in which apples must account for at least 1kg. What combination gives the lowest cost?

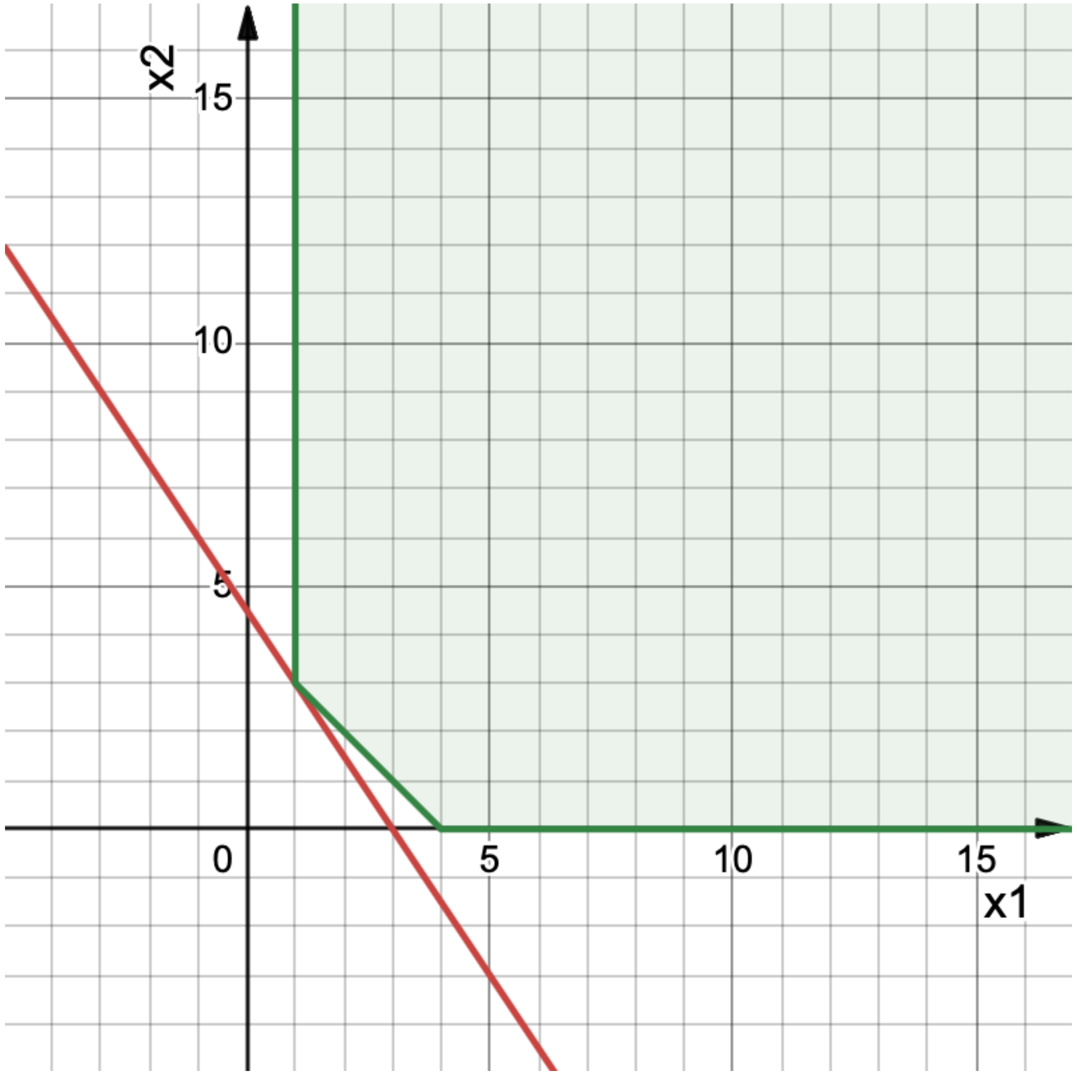
$$\begin{aligned} \text{minimize } & 3x_1 + 2x_2 \\ \text{subject to } & x_1 + x_2 \geq 4, \\ & x_1 \geq 1,\ x_2 \geq 0. \end{aligned}$$

Linear Programming (LP)

minimize $3x_1 + 2x_2$

subject to $x_1 + x_2 \geq 4,$

$x_1 \geq 1,\ x_2 \geq 0.$



Mixed Integer *LP* (MILP)

The objective function and constraints are all **linear**, plus **binary** decision variables.

$$
\begin{aligned}
\text{minimize } & f(x) = c^T x + d^T z \\
\text{subject to } & Ax + Bz \leq C, \\
& x \in \mathbb{R}^n, \\
& z \in \{0,1\}^k.
\end{aligned}
$$

Mixed Integer *LP* (MILP)

A factory can produce two products, A and B. Producing them requires paying fixed start-up costs (60 for A and 40 for B), and each unit produced generates profits (6 for A and 4 for B). Total machine capacity is limited to 20.

The decision is how much to produce of each product (continuous variables) and whether to activate production for each product (binary variables).

The goal is to maximize total profit.

Mixed Integer *LP* (MILP)

Let $x_1, x_2 \geq 0$ be production quantities, and $y_1, y_2 \in \{0,1\}$ be whether product A,B is produced. Then we have:

maximize $6x_1 + 4x_2 - 60y_1 - 40y_2$

subject to:

$x_1 + x_2 \leq 20$ (capacity constraint altogether),

$x_1 \leq 20y_1$ (capacity constraint for product A),

$x_2 \leq 20y_2$ (capacity constraint for product B),

$x_1, x_2 \geq 0$, $y_1, y_2 \in \{0,1\}$.

Non-linear Programming (NLP)

The objective function and constraints can be **non-linear**, including powers, exponentials, logarithms, integrals, products, etc.

$$\text{minimize } f(x)$$

$$\text{subject to } g_i(x) \leq 0.$$

We can use non-linear function to depict more realistic trends.

Non-linear Programming (NLP)

A 10-unit mixture must be made from two raw materials, A and B. Each material has a different unit cost (2 for A and 3 for B), and the mixture incurs an additional quality-penalty that depends quadratically on the proportions of A and B, given by one tenth the sum of raw materials amounts squared (i.e., $(A^2+B^2)/10$).

The decision is how much of A and B to mix (continuous variables).

The goal is to minimize total cost + nonlinear penalty.

Non-linear Programming (NLP)

Let $x, y \geq 0$ be amount of raw materials A and B in the 10-unit mixture. Then we have:

minimize $2x + 3y + \frac{1}{10}(x^2 + y^2)$

subject to:

$x + y = 10,$

$x, y \geq 0.$

Mixed Integer NLP (MINLP)

The objective function and constraints can be **non-linear**, including powers, exponentials, logarithms, integrals, products, etc. In addition, there are **binary** decision variables.

minimize $f(x)$

subject to $g_i(x) \leq 0.$

We can use MINLP to model various problems.

How to Solve?

Solvers (give exact solution)

LP: simplex, MILP: branch & bound, NLP: gradient descent, etc.

Heuristic Algorithms (cannot ensure global optimality, but fast)

Greedy algorithm

Local search / hill climbing

Simulated annealing

Genetic algorithm

Ant colony optimization

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Thank you